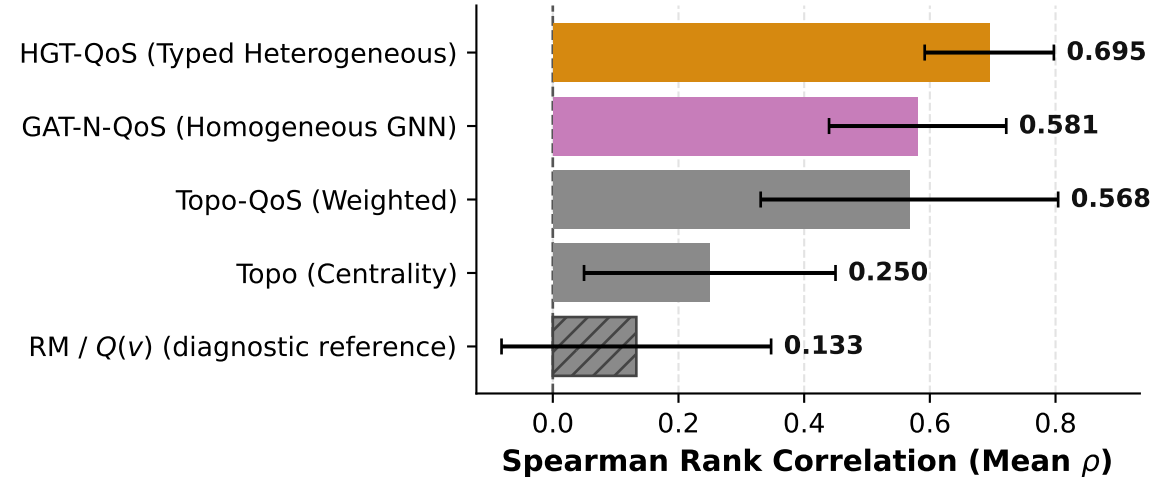
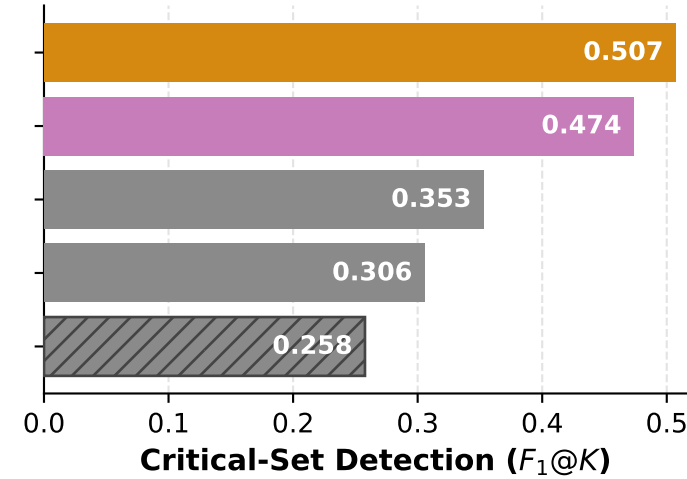


Software-as-a-Graph (SaG) Pre-Deployment Reliability Evaluation — application population

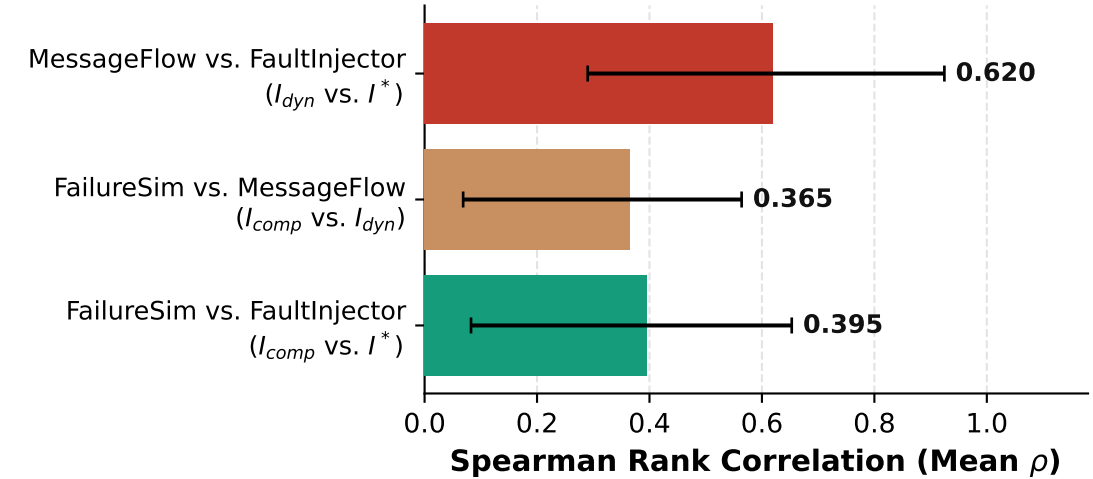
**A. Out-of-Distribution Generalization
(Leave-One-Scenario-Out ρ)**



**B. Top-K Critical Component Search
($F_1@K$, K = top 20%)**



**C. Simulation Oracle Agreement
(Rank Convergence & Range Across Scenarios)**



Mean top-K Jaccard overlap: 26.1%–36.0% (chance = 11.1%)
Read against the labeler's own seed-to-seed floor: 48%–100%